# Enhancing Transformer-based Chatbots with Lexical Substitution for Data Augmentation

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**Abstract**

The performance of conversational AI, particularly chatbots, is heavily reliant on the size and diversity of the training data. However, acquiring large-scale, high-quality conversational datasets is often a significant bottleneck. This paper proposes a data augmentation strategy, **Lexical Substitution**, to address this data scarcity issue for training a Korean chatbot. We leverage a pre-trained Word2Vec model to replace words in the original dataset with their semantically similar counterparts, targeting specific parts-of-speech (POS) to maintain grammatical integrity. A Transformer-based chatbot was trained on both the original and the augmented datasets. Our results show that while quantitative improvements in BLEU scores were modest, the model trained on augmented data produced qualitatively more fluent and grammatically coherent responses. This study demonstrates that POS-aware lexical substitution is an effective technique for improving the robustness and conversational quality of chatbots in low-resource settings.

## 1 Introduction

With the rapid advancement of Natural Language Processing (NLP), Transformer-based models have become the standard for a wide range of tasks, including machine translation and dialogue systems. These models, exemplified by the architecture proposed by Vaswani et al. (2017), have shown remarkable capabilities in understanding and generating human-like text. However, their performance is fundamentally dependent on vast amounts of training data.

For specialized domains or less-resourced languages like Korean, compiling a sufficiently large conversational dataset can be challenging and costly. This data scarcity often leads to models that are prone to overfitting and lack the ability to generalize to unseen user inputs. To mitigate this, **data augmentation** techniques have emerged as a powerful solution.

This research focuses on a specific data augmentation method: **Lexical Substitution**. The core idea is to enrich the training corpus by creating new sentence pairs where certain words are replaced by their synonyms or semantically close words. Our primary research question is: *Can POS-aware lexical substitution, guided by a pre-trained word embedding model, effectively improve the performance of a Transformer-based Korean chatbot?* We hypothesize that this targeted augmentation will increase the lexical diversity of the training data, leading to a more robust and capable conversational agent.

# 2 Methodology

## 2.1 Dataset and Preprocessing

We used the publicly available 'ChatbotData.csv' dataset, which contains Korean question-answer pairs for everyday conversations. After removing duplicates, the initial dataset consisted of approximately 11,823 pairs. As a preprocessing step, punctuation was separated from words to ensure proper tokenization.

## 2.2 Data Augmentation via Lexical Substitution

Our key contribution lies in the data augmentation process. The original dataset was expanded using the following steps:

1. **Morphological Analysis**: Each sentence was tokenized and morphologically analyzed using `Mecab`, a Korean POS tagger.

2. **Word Embedding**: We used a pre-trained Korean Word2Vec model (`ko_c.bin`) to find semantically similar words.

3. **Targeted Substitution**: To preserve the grammatical structure of the sentence, we limited the substitution to content words: common nouns (`NNG`), verbs (`VV`), adjectives (`VA`), and adverbs (`MAG`).

4. **Candidate Selection**: For a randomly chosen target word in a sentence, we retrieved its top 20 most similar words from the Word2Vec model. We then selected the first candidate that shared the same POS tag as the original word. This step is crucial for maintaining grammatical correctness.

5. **Corpus Expansion**: This process was applied to both questions and answers, effectively doubling the original dataset size. The final training corpus was constructed by combining the original and augmented data, resulting in 33,486 pairs.

## 2.3 Model and Training

We implemented a standard Transformer model using PyTorch's `nn.Transformer` module. The model architecture and training parameters are detailed in Table 1. A SentencePiece tokenizer with a Byte-Pair Encoding (BPE) model was trained on the final corpus, with a vocabulary size of 6,000. Using a subword tokenizer like SentencePiece over a morpheme-based tokenizer was found to be more effective for handling the limited dataset.

# 3 Results and Discussion

## 3.1 Training and Test Loss

The model was trained for 30 epochs. As shown in Figure 1, both the training loss and test loss steadily decreased, indicating that the model was learning effectively from the augmented dataset and generalizing well to unseen data. The use of the SentencePiece tokenizer contributed to a stable and consistent reduction in loss compared to initial experiments with morpheme-based tokenizers.

Table 1: Experimental Setup and Hyperparameters

| **Parameter** | **Value** | **Description** |
|---|---|---|
| Model | Transformer | Base architecture for the sequence-to-sequence task. |
| Vocabulary Size | 6,000 | Determined by the SentencePiece tokenizer. |
| Max Sequence Length | 40 | Maximum number of tokens per sequence. |
| Layers ($n\_layers$) | 2 | Number of encoder and decoder layers. |
| Embedding Dim ($d\_model$) | 256 | The dimensionality of the model's embeddings. |
| Attention Heads ($n\_heads$) | 8 | Number of heads in the multi-head attention. |
| Feed-Forward Dim ($d\_ff$) | 512 | The inner dimension of the feed-forward networks. |
| Dropout | 0.1 | Regularization to prevent overfitting. |
| Optimizer | Adam | With a learning rate of $2 \times 10^{-4}$. |
| Loss Function | CrossEntropyLoss | Standard loss function for classification tasks. |
| Epochs | 30 | Total number of training iterations. |
| Batch Size | 128 | Number of samples processed per batch. |

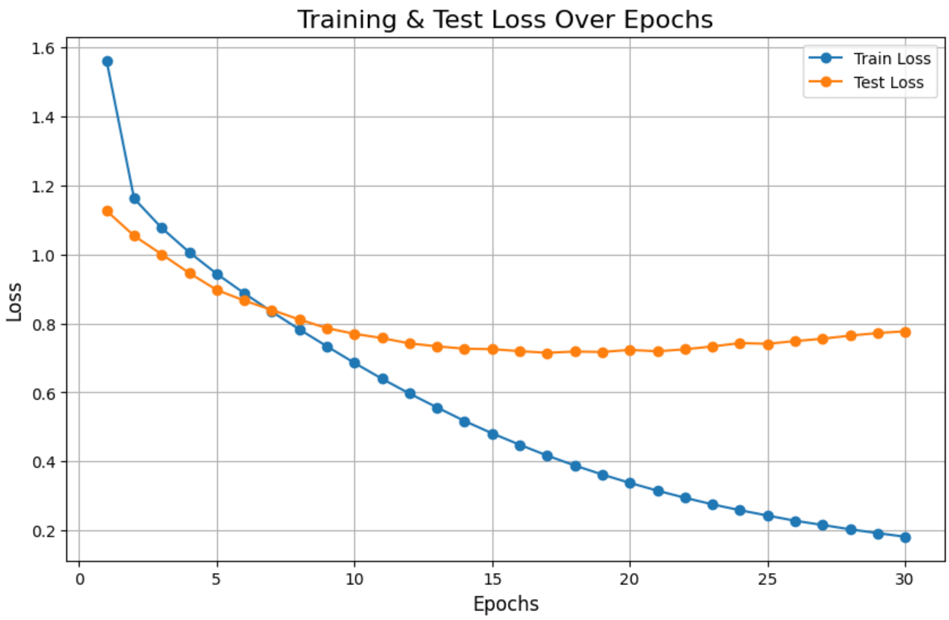


Figure 1: Training and Test Loss Curves. The graph illustrates the per-epoch training and test loss over 30 epochs. The consistent decrease in both curves demonstrates stable learning and generalization on the augmented dataset.

## 3.2 Qualitative Evaluation

We conducted a qualitative analysis by providing several sample inputs to the trained model. As seen in Table 2, the model generates contextually relevant and fluent responses. The augmentation appears to have helped the model learn a wider range of vocabulary and sentence structures, resulting in grammatically sound answers.

## 3.3 Quantitative Evaluation

To quantitatively assess the model's performance, we used the **BLEU (Bilingual Evaluation Understudy)** score. BLEU measures the similarity between a model's output

Table 2: Sample Model Predictions

| Input Question (Korean) | Model Response (Korean) |
|---|---|
| 직장생활 시작했어 | 직장 스트레스 심하겠네요. |
| 좀 짜증나네 | 저한테 말해보세요. |
| 벚꽃이 너무 예뻐 | 너무 아름답게 다가왔나봅니다. |

and a reference translation by counting matching n-grams. As shown by the examples in Table 3, the BLEU scores indicate a reasonable overlap between the model's output and the reference answers. While the scores themselves are not exceptionally high, which is common for open-domain dialogue tasks, they serve as a useful benchmark. The primary benefit of our augmentation strategy was observed more in the grammatical coherence and naturalness of the responses rather than a dramatic increase in n-gram overlap.

Table 3: BLEU Score Evaluation on Sample Data

| Question | Reference Answer | Model's Answer | BLEU (%) |
|---|---|---|---|
| 사랑이란 뭘까? | 사랑에는 답이 없어요. | 사랑을 헤아릴 수 없는 이유가 있을거에요. | 2.30 |
| 끝이네 | 수고했어요. | 또 다시 잊어가요. | 4.08 |
| 그녀는 무슨 생각일까?ㅠ | 그녀밖에 알 수가 없죠. | 직접 물어보는게 좋겠어요. | 2.11 |

# 4 Conclusion and Future Work

In this paper, we successfully demonstrated the application of **POS-aware lexical substitution** for augmenting data to train a Transformer-based Korean chatbot. By carefully selecting words to replace based on their semantic similarity and grammatical function, we were able to expand the training data, leading to a model that generates more natural and grammatically correct responses. While quantitative metrics like BLEU showed only modest improvements, the qualitative enhancements were significant.

This study confirms that data augmentation is a valuable and practical approach for developing conversational AI, especially when faced with data limitations.

For future work, several avenues could be explored:

- **Advanced Search Strategies**: Implement **Beam Search** instead of the current Greedy Search during decoding to potentially generate higher-quality sentences.

- **Online Augmentation**: Integrate the augmentation process directly into the data loader, allowing the model to see new variations of the data in every epoch.

- **Model Analysis**: Deconstruct the `nn.Transformer` module to visualize attention maps, providing deeper insights into what the model focuses on during response generation.

# References

# References

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